Mini Project
AUG 2025 to DEC 2025

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Title of the pre-final year BE Mini project work (V Sem)	:	**Real-Time Detection of Forest Fires Using FireNet-CNN and Explainable AI Techniques**		
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Field/Domain/Area of project	:	Machine Learning		
Project Synopsis Approved by Guide/Proj Coord/HoD	:	**Guide** Yes / No	**Proj Coord** Yes / No	**HoD** Yes / No

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Title: Real-Time Detection of Forest Fires Using FireNet-CNN and Explainable AI Techniques

Introduction:

1. Wildfires are an increasing global threat, causing severe ecological and economic damage.
2. Traditional automated fire detection relies on static image analysis using Convolutional Neural Networks (CNNs).
3. These CNN-based approaches face high false positive rates from fire-like phenomena such as clouds, fog, or sun glare.
4. They lack temporal understanding, limiting their ability to distinguish dynamic fire events from static visuals.
5. Many models function as "black boxes," making their predictions hard to trust in critical situations.
6. FireNet 2.0 addresses these challenges through spatio-temporal analysis, multimodal data fusion, and integrated explainability.

Objective and Scope:

1. To develop a spatio-temporal deep learning model using a ConvLSTM architecture to analyze video sequences, enabling it to distinguish between persistent fire events and transient non-fire phenomena.
2. To design and implement a multimodal fusion system that processes both RGB (visual) and simulated thermal video streams, enhancing detection robustness under varied environmental conditions.
3. To integrate an uncertainty estimation mechanism using Evidential Learning, allowing the model to provide a confidence score with its predictions to reduce false alarms.
4. To incorporate Explainable AI (XAI) techniques, specifically Grad-CAM, to generate visual heatmaps that highlight the key features influencing the model's decisions, thereby improving transparency and operator trust.

Literature Survey:

S.No	Name of the article	Author's Name	Publication Year	Pros	Cons	Gap Analysis
1.	**Forest Fire Detection Algorithm Based on Improved YOLOv11n**	Kangqian Zhou and Shuihai Jiang	2025	Achieves high accuracy with lower computational complexity and faster inference speed,	Still faces challenges from diverse targets and complex backgrounds that may occur in different fire scenario	More data augmentation techniques and model adaptation optimization methods needed for diverse targets and complex backgrounds.
2.	**Real-Time Detection of Forest Fires Using FireNet-CNN and Explainable AI Techniques**	Gazi Mohammad Imdadul Alam et al.	2025	It achieves high accuracy, precision, and recall; optimized for resource-constrained devices.	Relies on synthetic data, The datasets used have class imbalance issues; requires extensive real-world validation	Addresses limitations of existing methods by proposing a custom, lightweight CNN with integrated XAI and diffusion model-based data augmentation

3.	**Hybrid Backbone-Based Deep Learning Model for Early Detection of Forest Fire Smoke**	Gökalp Çınarer	2025	Deep learning approach for high-accuracy fire signature detection from smoke.	Ability to distinguish phenomena may weaken in dark, hazy, rainy, and smoggy weather.	Integrate different image processing and smoke detection techniques.
4.	**Real-Time Fire Object Detection System Using Machine Learning**	Venkata Bhargavi Akuthota et al.	2025	Employs ML algorithms for enhanced early detection Designed for alerting emergency responders	It often fail in accuracy and response time in complex settings; prior work largely uses still images, indicating a need for temporal analysis.	Can Overcome by integrating sophisticated deep learning and computer vision, real-time fire object detection and improved early warning systems
5.	**Forest Fires in India: A Review**	Reuben Sewak, Mehak Vashisth, Lovleen Gupta	2021	Review of forest fire conditions in India; identifies vulnerable	Ground-based detection systems are under-resourced,	An adaptive, long-term fire management plan is needed.

				clusters and impacts on biodiversity, climate, and soil; outlines government strategies.	with poorly trained and unpaid staff.	Accounting of the economic costs and benefits of forest fires is needed
6.	**Convolution LSTM Network: A ML Approach for Precipitation Nowcasting**	Xingjian Shi et al.	2015	ConvLSTM outperform in capturing spatio-temporal correlations, handling boundary conditions	Blurring effect in longer-term predictions due to inherent uncertainity	Enable prediction of fire dynamics across consecutive time steps, in any environment than CNN.

Proposed Methodology:

Our approach is to build a comprehensive system in three stages. First, we will construct a robust data pipeline that generates and processes multimodal video data. Second, we will develop the core FireNet 2.0 model, which features a dual-backbone CNN for feature extraction from RGB and thermal streams, a ConvLSTM layer for temporal analysis, and an evidential output head for uncertainty prediction. Finally, we will integrate a Grad-CAM module to provide visual explanations for the model's predictions.

- **Tools, techniques and technologies we plan to use:**
  - Programming Language: Python
  - Deep Learning Framework: PyTorch
  - Core Libraries: OpenCV (for video processing), NumPy (for numerical operations), Matplotlib (for visualization), Scikit-learn (for evaluation metrics).
  - Development Environment: Jupyter Notebook / Google Colab with GPU support.

- **Work flow plan**:
  a. Data Acquisition & Augmentation
  b. Image Preprocessing & Segmentation
  c. Model Development
  d. Training & Validation
  e. Explainable AI Integration
  f. System Deployment & Benchmarking

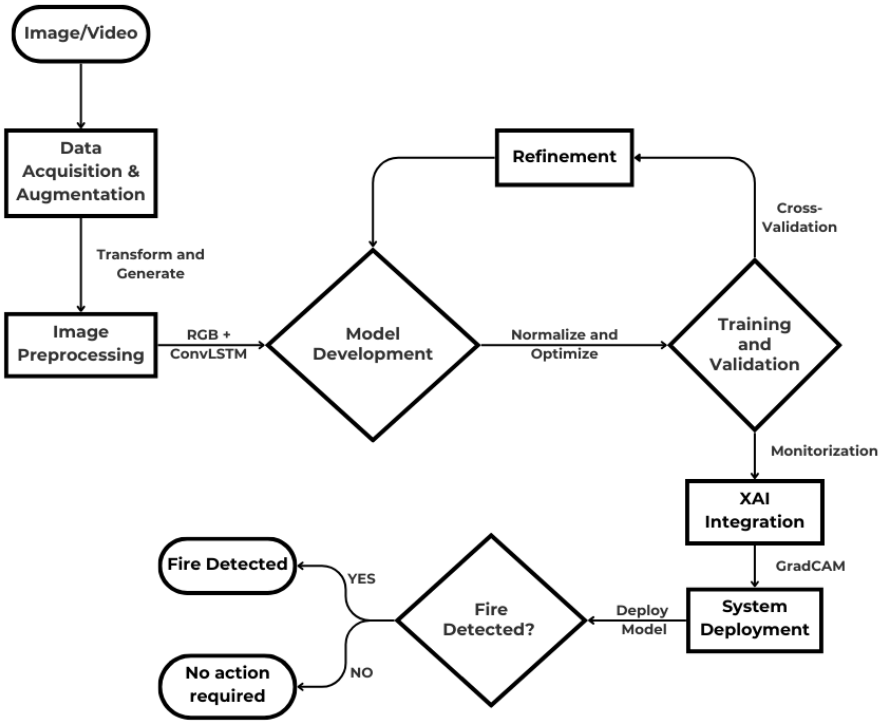
Here is the tabular column summarizing the models used in the project, corresponding datasets, performance metrics (accuracy and precision), use cases, and the innovations integrated along with the project:

ML Models to be used in this Project:

Model/ Innovation	Dataset Used	Precision	Accuracy	Use Cases
FireNet-CNN	Dataset A and Dataset B	99.41	99.05	Custom CNN optimized for forest fire detection using image data, balancing accuracy and computational efficiency.
VGG16 (Pre-trained)	Dataset A and Dataset B	~95.30	Lower than FireNet	Standard pre-trained CNN architecture used for benchmarking fire image classification.
VGG19 (Pre-trained)	Dataset A and Dataset B	~95.24	Lower than FireNet	Deeper variant of VGG16, used for performance comparison in fire detection.
Inception V3 (Pre-trained)	Dataset A and Dataset B	~97.53 (A), 98.88 (B)	Slightly lower than FireNet	Deep CNN with inception modules, shows strong performance in image classification tasks.

YOLO V11(Fine-tuned)	Fire image datasets, open fire/video datasets	~95.33	~99.5	Real-time object detection & fire localization from camera feed; fast response in surveillance/monitoring

Block Diagram:



Software and Hardware Requirements:

- **Hardware:**
  - A computer with a modern multi-core CPU.
  - Minimum 8 GB of RAM.
  - An NVIDIA GPU with CUDA support (e.g., GTX 1650, RTX 2060 or higher) is highly recommended for feasible training times.
- **Software:**
  - Operating System: Windows, macOS, or Linux.
  - VS Code, Jupyter Notebook, Python 3.8+

- PyTorch 1.10+
- Python Libraries: OpenCV, NumPy, Matplotlib,Keras,Tensorflow,Scikit Learn

Applications:

The advanced capabilities of FireNet 2.0 enable a wide range of practical applications beyond simple fire detection, creating a comprehensive monitoring and response ecosystem.

- Real-Time Monitoring and Early Warning Systems
- Dynamic Fire Behavior and Spread Analysis.
- Intelligent Decision Support for Command Centers.
- At-the-Edge Deployment on Autonomous Systems.
- Post-Incident Analysis and Forensic Investigation.

Advantages :

The innovative architecture of FireNet 2.0 provides significant advantages over existing fire detection systems, addressing their core weaknesses in reliability, trust, and performance.

- Drastically Reduced False Alarms.
- Enhanced Operator Trust and Actionable Insights.
- Superior Robustness and All-Weather Performance.
- High Scalability and Efficiency for Real-World Deployment.
- Continuous Improvement and Domain Adaptability.

Expected Outcome:

- Real-time forest fire detection system with high accuracy (typically >95%) and rapid inference times suitable for field applications.
- Transparent and explainable AI outputs that highlight model reasoning for each detection, fostering trust in system alerts.
- Improved robustness and generalization through large-scale data augmentation and balanced datasets.
- Deployable prototype on affordable hardware such as webcams or edge devices, demonstrating real-world viability.

- It can able to distinguish Fire or Smoke or non-fire elements, predict where the next surrounding gets affected by the fire or not

References:

[1] "Forest Fires in India: A Review", Reuben Sewak1, Mehak Vashisth1, Lovleen Gupta1 Year: 2021

[2] "Real-Time Fire Object Detection System Using Machine Learning" -Venkata Bhargavi. Akuthota , Khadar Basha Syed, Dhanush Ramineni , and Vikram Guduru, Department of CSE, Vignan's Foundation for Science, Technology and Research Vadlamudi, Guntur, Andhra Pradesh, India

[3] "Hybrid Backbone-Based Deep Learning Model for Early Detection of Forest Fire Smoke", Gökalp Çınarer Year: 2025

[4] "Forest Fire Detection Algorithm Based on Improved YOLOv11n", Kangqian Zhou and Shuihai Jiang School of Mechanical and Electronic Engineering, Nanjing Forestry University, Nanjing 210037, China; Institute of Intelligent Control and Robotics Technology, Nanjing Forestry University, Nanjing 210037, China Year: 2025

[5] "Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting", Xingjian Shi , Zhourong Chen , Hao Wang, Dit-Yan Yeung Department of Computer Science and Engineering, Hong Kong University of Science and Technology Year:2015

[6] "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization", Ramprasaath R. Selvaraju , Michael Cogswell , Abhishek Das , Ramakrishna Vedantam , Devi Parikh , Dhruv Batra Year: 2019

[7] Real-Time Detection of Forest Fires Using FireNet-CNN and Explainable AI Techniques Gazi Mohammad Imdadul Alam , Naima Tasnia, Tapu Biswas, MD Jakir Hossen (Senior Member, IEEE), Sharia Arfin Tanim, and MD Saef Ullah Miah Year: 2025

Guide Signature | Project Coordinator Signature | HOD Signature